

Glass Ionomer Cement

outlines

- Introduction
- Composition
- Classification
- Advantage and Disadvantage
- Indication and contraindication
- Tooth preparation steps
- Mixing of GI
- Modifications of GIC
- conclusion

- Adhesion of restorative materials to tooth structure is an important objective.
- It is always required that a restoration should resemble the tooth in all respects.
- It should have identical properties and should bond to the surrounding enamel and dentin.
- The glass ionomer cements are one of the products developed to meet all these criteria.

- Glass Ionomer Cement was introduced to dentistry in 1972 by **Wilson and Kent**
- GIC is basically formed by the reaction between **polyalkenoic acid** and **alumino-silicate glass**
- Extensive use of this cement to replace dentin, has given it different names: **Dentin substitute, man-made dentin and artificial dentin.**

CLASSIFICATION OF GLASS IONOMER CEMENTS

Classification of GICs according to their use

Type I—For luting cements

Type II—For restorations

Type III—Liners and bases

Type IV—Fissure sealants

Type V—Orthodontic cements

Type VI—Core build up.

Newer classification

Type I—Luting cements

Type II—Restorative esthetic or reinforced cements

Type III—Liner or base

COMPOSITION OF GLASS IONOMER CEMENT

1. Powder

- Acid soluble calcium fluoroaluminosilicate glass.
- Silica, Alumina, Aluminum fluoride, Calcium fluoride, Sodium fluoride, Aluminum phosphate

2. Liquid

- ✓ Polyacrylic acid
- ✓ Tartaric acid:
- ✓ Water : Medium of reaction & hydrates the reaction products

SETTING REACTION

- Basically in autocure glass ionomer cement, setting reaction is an acid-base reaction between the acidic polyelectrolyte and the aluminosilicate glass.
- It occurs in three different but overlapping stages
 - ✓ Ion-leaching phase
 - ✓ Hydrogel phase
 - ✓ Polysalt gel phase.

- **Three stages of GIC setting reaction:**

- 1. Ion-leaching phase**

- ✓ When powder and liquid are mixed, polyacid attacks the glass particles (called leaching) to release Ca^{2+} and Al^{3+}

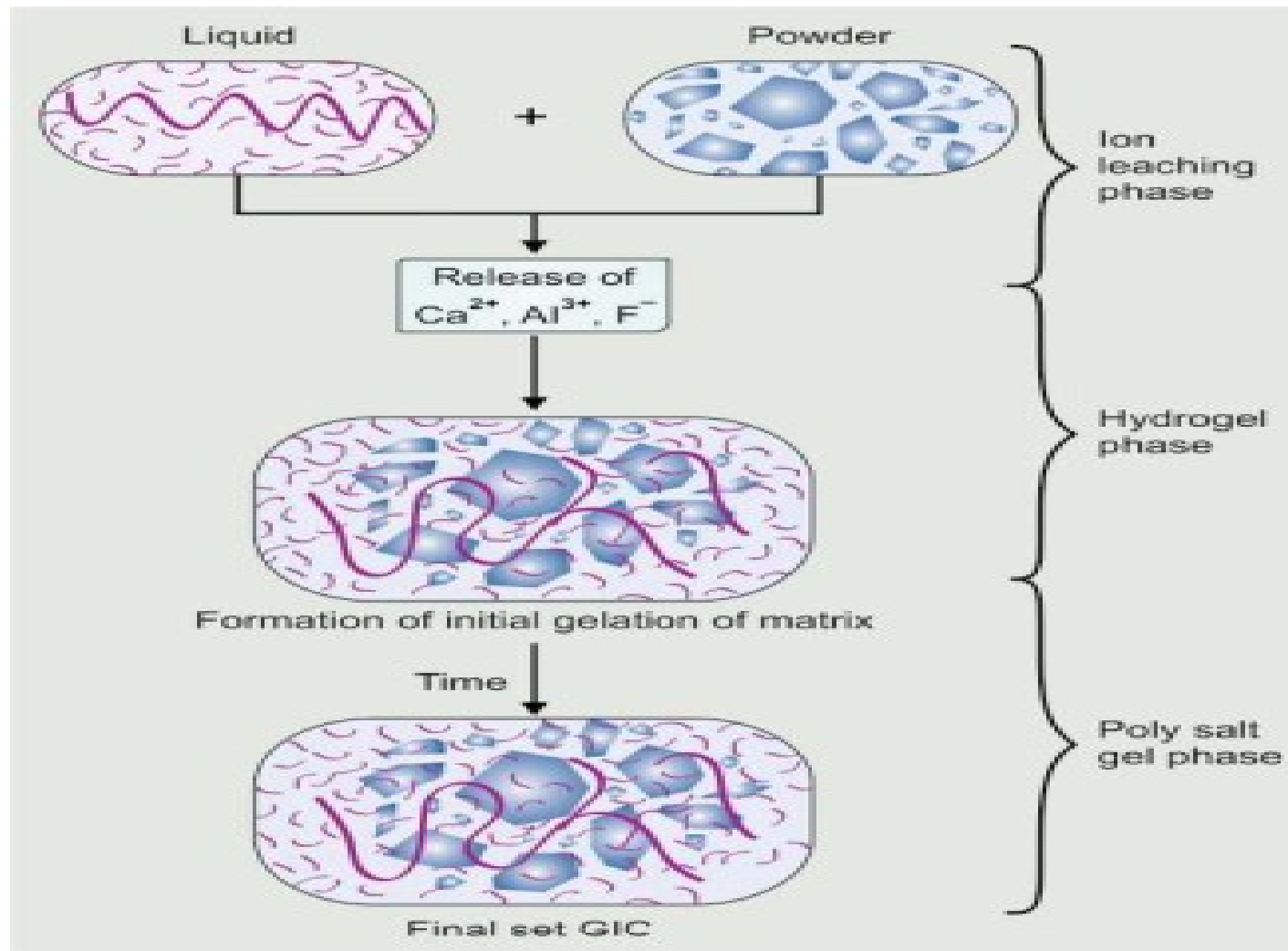
- 2. *Hydrogel Phase***

- ✓ In this phase, calcium ions are released rapidly.
- ✓ These liberated ions cross-link with polyacrylic acid, i.e. calcium bridges to form a calcium polycarboxylate gel in which the non-reacted glass is embedded.
- ✓ This reaction is initial setting of cement.

3. Polysalt Gel Phase

- ✓ continued attack of hydrogen ions causes delayed release of Al ions from silicate glass in form of AlF ions which are deposited in the already preformed matrix to form a water-insoluble Ca-Al-Carboxylate gel.
- ✓ Aluminum ions are responsible for providing strength to cement.

SETTING REACTION OF GIC

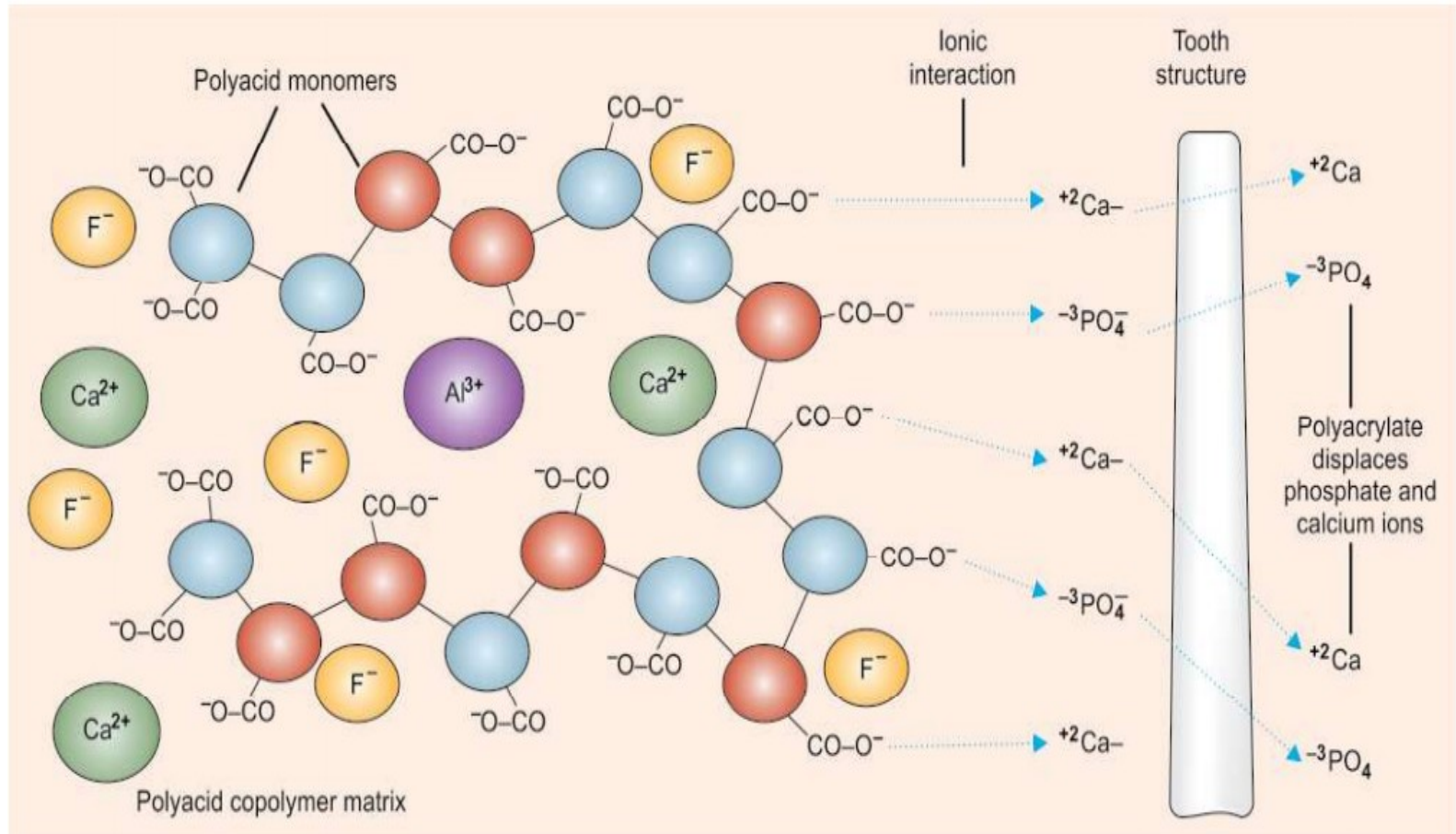


➤ GIC is

- ✓ self-cured
- ✓ tooth colored
- ✓ fluoride releasing
- ✓ bond to tooth structure without an additional bonding agent.

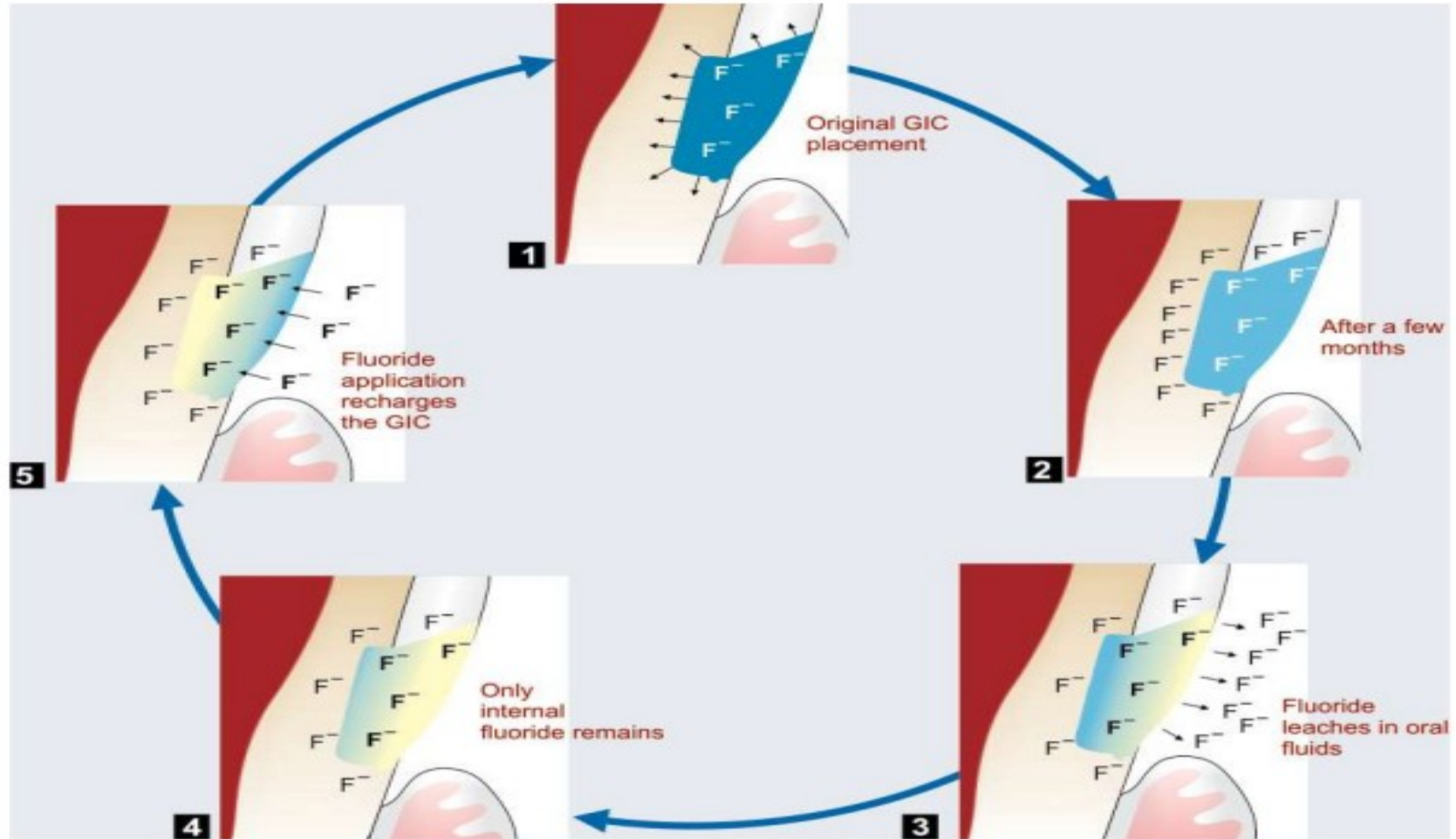
- ✓ GIC adheres chemically to enamel and dentine.
- ✓ Since it links to calcium ions in the dental tissues, the bond to enamel is stronger than the bond to dentine because enamel is more highly mineralized.
- ✓ GICs are appropriate for restoring cavities where little or no cavity preparation is undertaken.

Chemical bonding of GIC to tooth structure



- Glass ionomers possess the quality of releasing fluoride when exposed to the oral environment.
- This property may render glass-ionomer restorations more resistant to recurrent caries.
- Because of this anticariogenic quality, glass ionomer may be the material of choice for restoring root-surface caries in patients with high caries activity and where esthetics is not critical.

Diagram showing reservoir effect of glass ionomers: (1) Fluoride ions leach out from GIC into tooth; (2) Balance of fluoride between tooth and restoration; (3) Fluoride enters in saliva; (4) Both tooth and restoration are short of fluoride; (5) Topical application of fluoride replenish the cement



USES OF GLASS IONOMER CEMENTS

- **As Pit and Fissure Sealants**
- **As Liners and Bases**
- **As Luting Agents**
- **As Orthodontic Brackets Adhesives**
- **For Restorations of Class III and Class V Lesions**
- **Restoration of Root Caries**
- **High Caries Risk Patients**
- **Emergency Temporary Restorations**
- **For Intermediate Restorations**
- **For Core Build Up**
- **Atraumatic Restorative Treatment (ART)**

contraindication

- Stress bearing areas in permanent teeth: class I,II and IV carious lesions or fractured incisors.
- Lost cusp areas
- Restorations involving large labial surface where esthetics is of major importance
- Mouth breather and patients with xerostomia

CLINICAL STEPS FOR PLACEMENT OF GLASS IONOMER CEMENT

- Isolation
- Instrumentation and tooth preparation
- Mixing of GIC
- Restoration
- Finishing and polishing
- Surface protection.

Isolation

- Saliva control is important for successful glass ionomer restorations.
- If moisture contaminates the cement during manipulation and setting, the gel will weaken and wash out prematurely.
- Commonly used methods for isolation are rubber dam, retraction cords, cotton rolls and saliva ejectors

Tooth Preparation

- Tooth preparation for glass ionomer cement is done in two ways:
 1. Mechanical preparation
 2. Chemical preparation (conditioning)

Mechanical preparation for GLC restoration

- **Outline form**

- ✓ Outline form is governed by caries, discoloration and esthetics.
- ✓ Cavity preparation is done very conservatively; the main points are as follows.
 - ✓ contact areas are not cut to bring the margins to self-cleansing areas, unless not involved in caries,
 - ✓ The unsupported enamel is not removed if it is not exposed to heavy masticatory forces

B. Resistance form

- ✓ The material should be sufficiently thick in bulk, at least
 - ✓ 1mm thick to resist the fracture under light masticatory load and
 - ✓ 2-3 mms thick in the areas where masticatory load is heavy.

C. Retention form:

- ✓ GIC has true chemical adhesion to enamel and dentine.
- ✓ Hence, excessive major and undue undercuts , dovetails and retention groove are not required at the cost of healthy tooth structure.

D. (Chemical preparation) Debridement

- ✓ **GIC chemically bonds to the cavity walls; hence, the cavity wall must be clean , disinfected and conditioned.**
- ✓ **To etch the dentin, 10% polyacrylic acid is placed in the preparation for approximately 20 seconds, followed by rinsing and removal of excess water, leaving the dentin moist**
- ✓ **Following preparation of the tooth, deeply excavated areas (within 0.5 mm of the pulp) may need protection with a calcium hydroxide liner.**

Mixing of Glass Ionomer cement

- ✓ Mixing should be done using the powder: liquid ratio as recommended by the manufacturer.
- ✓ Mixing should be done:
- ✓ On a cool and dry glass slab or paper pad With a help of flat and firm plastic spatula - glass slab is better than paper pad for mixing as it can be cooled to prolong the working time if required.
- ✓ Liquid should not stay on paper pad longer than 1minute (some of it may soak into it)

➤ Correct mixing of GIC is critical for clinical success

- ✓ A high powder: liquid ratio gives increased strength to the cement and its adhesive interface.
- ✓ Reducing the amount of powder in the mix will lead to a cement that sets by 'drying out' rather than the correct chemical reaction causing:
 - Reducing strength
 - Poor appearance
 - Questionable dimensional stability

Mixing of the cement

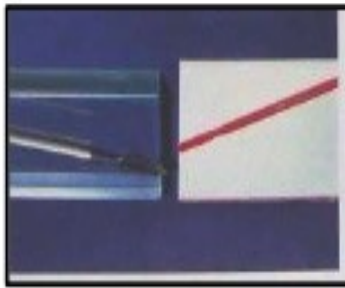


Full spoon, no excess



Tip liquid bottle to side, then
invert completely
If water / tartaric acid, only 1 drop
used.





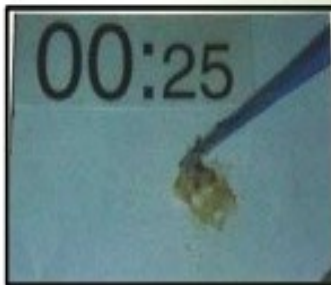
Liquid should not stay on paper pad longer than 1minute (some of it may soak into it)



Don't mix beyond 30 seconds
The objective is – only wet the particle – no dissolving it.



First half folded into liquid in 10-15seconds



Second half incorporated in 15 seconds
Small mixing area

Restoration

- ✓ **Adhesion to Tooth Structure**
- GIC adhere chemically by ion exchange with tooth structure
- Gross excess is quickly removed
- ✓ **Polymerization and Setting Reaction**
- Complex acid-base reaction occurs between the glass powder and the ionic polymer.
- Light activated polymerization - Light cure cement is photo activated for **60 seconds**.

Finishing and Polishing

- ✓ Excess is trimmed with sharp hand instruments.
- ✓ Finishing diamond points at very slow speed with petroleum lubricant are used without water coolant.
- ✓ The restoration is finally covered with colorless varnish or petrolatum jelly to prevent dehydration.
- ✓ Final finishing is done after 24 hours using coarse to fine finishing diamond points, abrasive strips.
- ✓ Dry finishing with abrasives will dehydrate the surface and make it porous resulting in early disintegration of the cement.
- ✓ After final finishing the surface is covered with varnish.

Atraumatic Restorative treatment (ART)

- ✓ In this technique the carious tooth tissue is removed with the help of excavators and hand instruments
- ✓ This is an effort to **make the filling cheaper and also available at the door steps** of the poor rural masses living in remote areas.
- ✓ Taking the advantages of **chemical adhesion and continuous fluoride release**
- ✓ The **glass ionomer cement filling is done without proper cavity preparation.**

- ✓ This technique is advantageous to patients as it is:
 - A. Less costly
 - B. Less time consuming
 - C. Can be used in remote rural areas where electricity is not available and as no costly dental instruments and equipment or any devices which are run by electricity are required.
 - D. It can be done without dental chair

Sandwich Technique

- Term “sandwich technique” refers to a laminated restoration using glass ionomer to replace dentin and composite resin to replace enamel
- Composite resin bonds micromechanically to the set glass ionomer and chemically to the HEMA in resin-modified glass ionomers
- *Synonyms of Sandwich Technique*
 - Replacement dentin technique
 - Bilayered technique
 - Laminate restoration technique.

➤ *Indications of Sandwich Technique*

- Large Class III, IV, V and class I and II lesions
- For laminate veneers.

Steps of Sandwich Technique

- Isolate the tooth to be prepared
- Prepare the tooth- Bevel the enamel margins to increase the composite resin bonding.
- Condition the prepared tooth using polyacrylic acid for optimal adhesion of GIC.
- Place freshly mixed fast setting GIC in the prepared tooth
- It is only necessary to etch a GIC with acid if the restoration has been in place for some time and has fully matured
- Apply bonding agent and cure it
- Place composite and cure in usual manner
- Do finishing and polishing of the restoration and finally recure it for 20 seconds

Modifications of GIC

❖ **Resin Modified Glass Ionomer /RMGI /**

- It consists of 80 % GIC and 20% visible light hardened resin components.
- In this system, powder is fluorosilicate glass with photoinitiator or chemical initiator. The liquid contains 15 to 25 percent resin component in the form of HEMA, a polyacrylic acid copolymer along with photoinitiator and water.
- As compared to traditional/conventional GIC – **Resin Modified Glass Ionomer** has
 - Greater strength
 - Less moisture sensitivity
 - Less soluble
 - Immediate finishing
 - Esthetics of finished restorations is similar to that of composites

❖ Compomer - Polyacid - Modified composite Resin (PMC) or Glass Ionomer resin Composite

- ✓ Compomer is contraction form of *Composite* and Glass-ion*omer*
- ✓ Compomers **have properties that are intermediate to** those of **Resin-modified glass ionomers** but are **closer to composites**.
- ✓ Setting reaction is **light activated**.
- ✓ Adhesive system of compomer is based on acid etch found with all composite resin.
- ✓ No chemical bond to tooth structure

❑ **Advantages of Compomer**

- Easy to place and polish
- Some fluoride release
- More esthetic than glass ionomer
- Better mechanical properties than glass ionomer

❑ **Disadvantages of Compomer**

- Inferior mechanical properties compared to composite
- Less fluoride release than glass ionomer
- No chemical bond to tooth structure

❖ Metal Modified Glass Ionomer Cement

- Glass ionomer have been modified by addition of filler particles ,to improve strength ,fracture toughness & resistance to wear.

- ✓ Improved handling
- ✓ Improved radiopacity
- ✓ Fluoride release –similar or slightly less than conventional
- ✓ Improved wear resistance

❖ Silver alloy admix / miracle mix

- ✓ Amalgam alloy mixed with Glass Ionomer powder
- ✓ This is made by mixing of spherical silver amalgam alloy powder with glass ionomer powder.

❖ Cerment

- ✓ Bonding of silver particles to glass ionomer particles by fusion through high temperature sintering.

Conclusion

- GI is a dental material used for filling and luting cement
- Bond chemically with tooth structure
- Release fluoride
- Prevent caries

➤ Reference

- Text book of operative dentistry 3rd edition
- Art and science of operative dentistry 7th edition

Thank you all